

UGANDA CHRISTIAN UNIVERSITY

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Faculty of Engineering, Design and Technology

Department of Computing and Technology

Digital Audio Communication Simulator: End-to-End System Simulation

Smart Digital Audio Communication Simulator: End-to-End System Simulation and Performance Analysis

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Abstract

This report details the design, implementation, and performance evaluation of a comprehensive end-to-end digital communication simulator developed in MATLAB. The system models a complete communication chain spanning analog message generation, uniform quantization, Pulse Code Modulation (PCM) encoding, polar NRZ line coding, digital modulation (BPSK/QPSK), Additive White Gaussian Noise (AWGN) channel transmission, demodulation, decoding, and signal reconstruction. Performance is rigorously analyzed across a signal-to-noise ratio (SNR) sweep from 0 dB to 20 dB, evaluating Bit Error Rate (BER), reconstruction Mean Squared Error (MSE), and Shannon channel capacity limits.

1 Introduction and System Architecture

The objective of this project is to build and evaluate a complete simulated digital communication chain. The modular software architecture is organized around a 6-stage pipeline handled by a master script (‘main.m‘) and core helper functions (‘functions/‘).

1.1 System Block Diagram

Below is the architectural block diagram representing the complete end-to-end simulation pipeline:

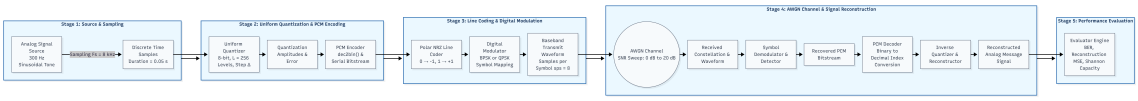


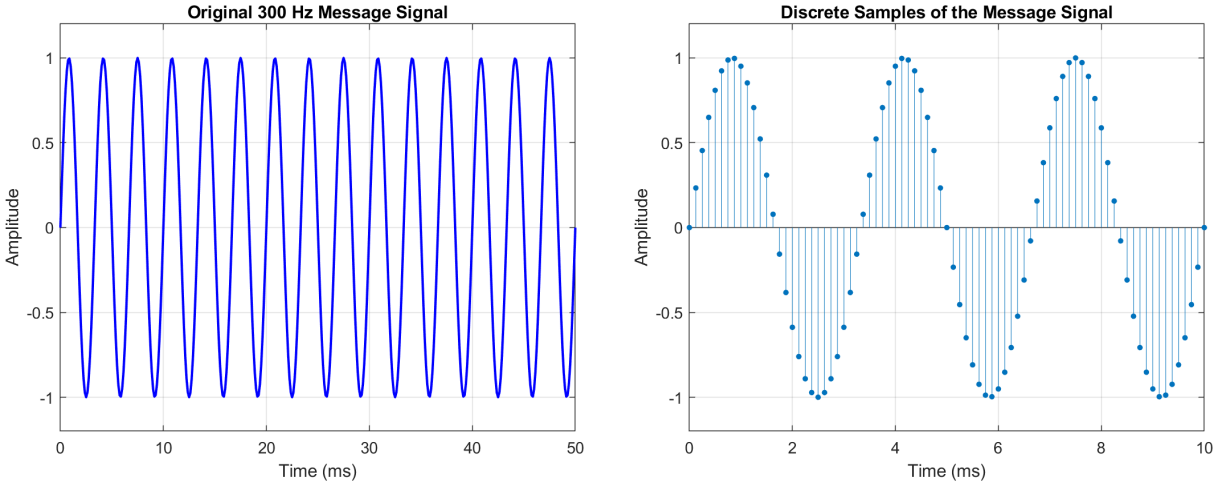
Figure 1: Complete End-to-End Digital Communication System Block Diagram.

2 Stage-by-Stage Simulation and Visualizations

The simulation executes the following sequential stages, with visual outputs captured at each step:

2.1 Stage 1: Source Generation and Sampling

A continuous baseband sinusoidal test tone ($f_m = 300\,\text{Hz}$) is generated with a sampling frequency of $F_s = 8\,\text{kHz}$ over a duration of 0.05 seconds.



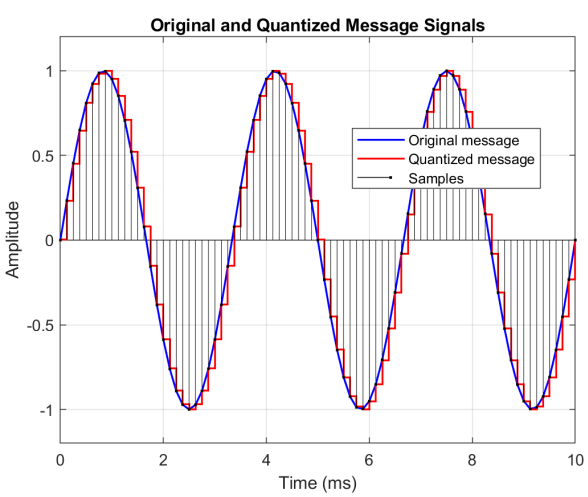
(a) Continuous message signal (300 Hz)

(b) Discrete samples ($F_s = 8\,\text{kHz}$)

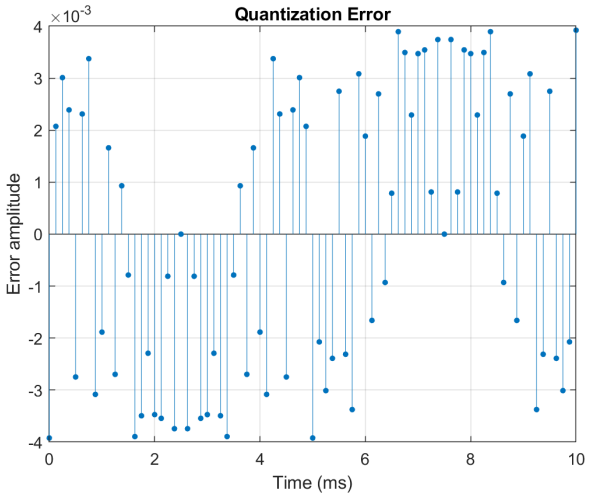
Figure 2: Source generation and sampling plots.

2.2 Stage 2: Quantization and PCM Encoding

The sampled signal undergoes 8-bit uniform quantization ($L = 256$ levels). Quantization indices are converted into a serial binary PCM bitstream and analyzed via error distributions.



(a) Quantized vs. Original waveform



(b) Quantization error distribution

Figure 3: Quantization stage outputs.

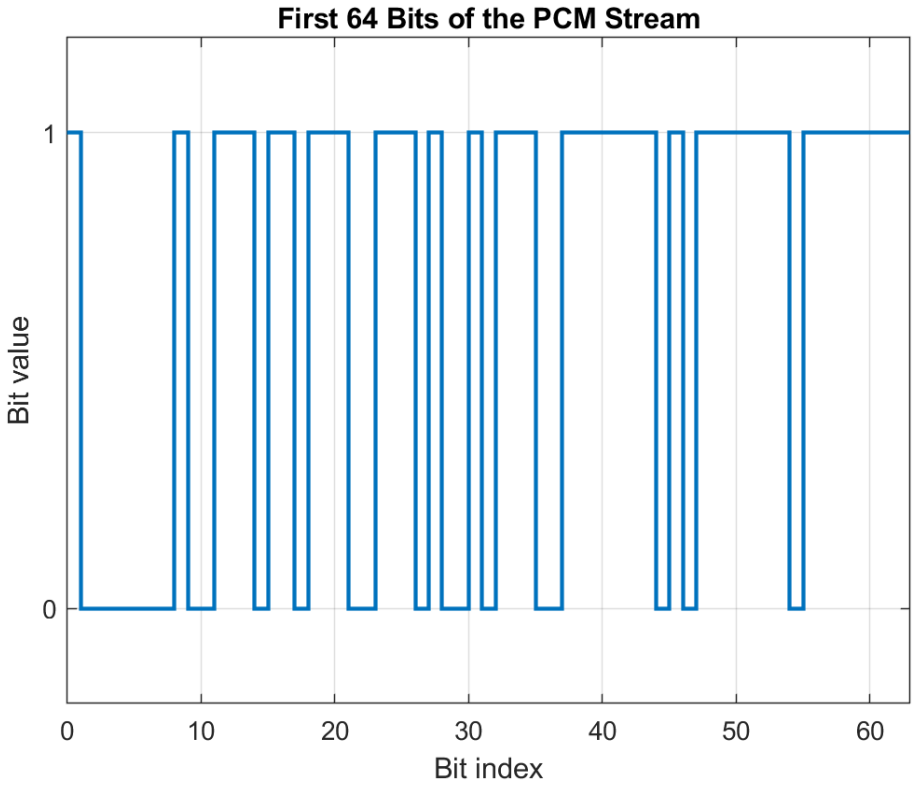
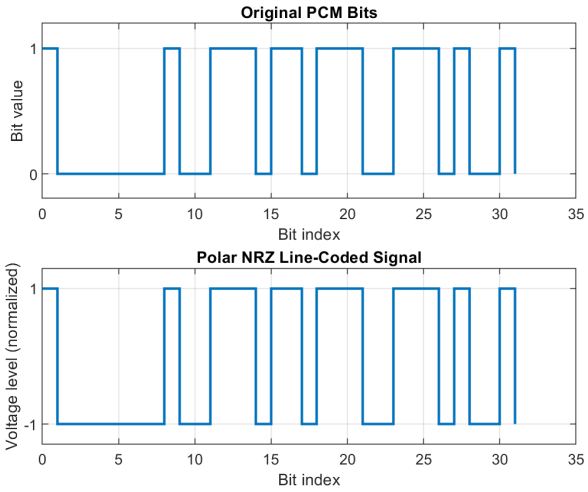


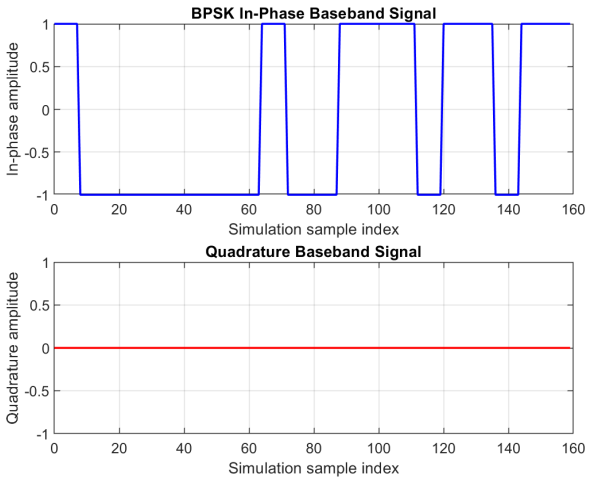
Figure 4: First 64 bits of the generated PCM serial bitstream.

2.3 Stage 3: Line Coding and Digital Modulation

The PCM bitstream is converted into polar Non-Return-to-Zero (NRZ) line codes, modulated into baseband waveforms, and mapped onto complex constellation symbols (BPSK/QPSK).



(a) Polar NRZ waveform



(b) Modulated baseband waveform

Figure 5: Line coding and baseband modulation outputs.

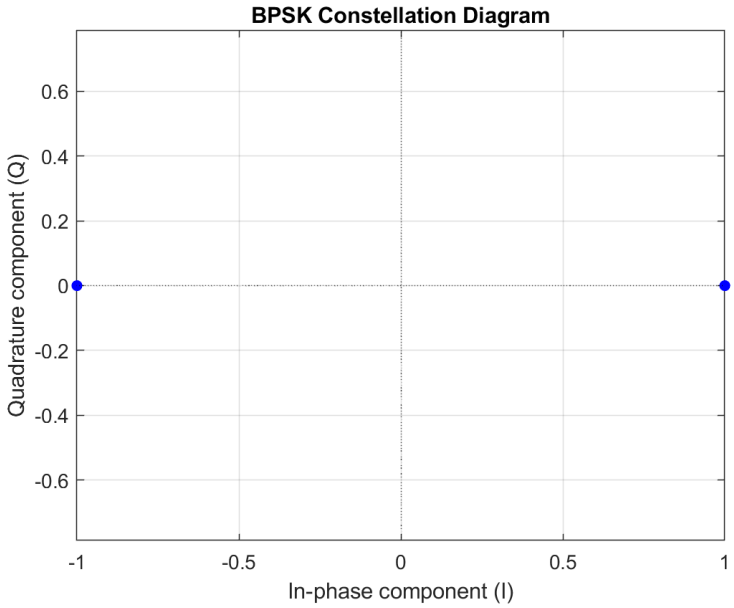
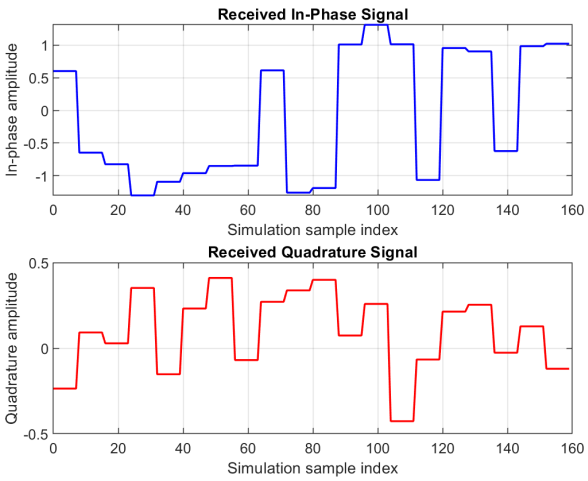


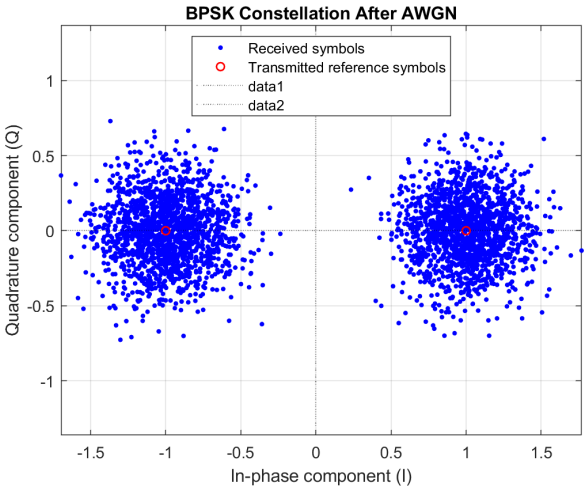
Figure 6: BPSK/QPSK modulation constellation diagram.

2.4 Stage 4: AWGN Channel, Demodulation, and Reconstruction

The modulated symbols are transmitted across an AWGN channel. At the receiver, symbols are demodulated, bitstreams are recovered, and the signal is inverse-quantized.

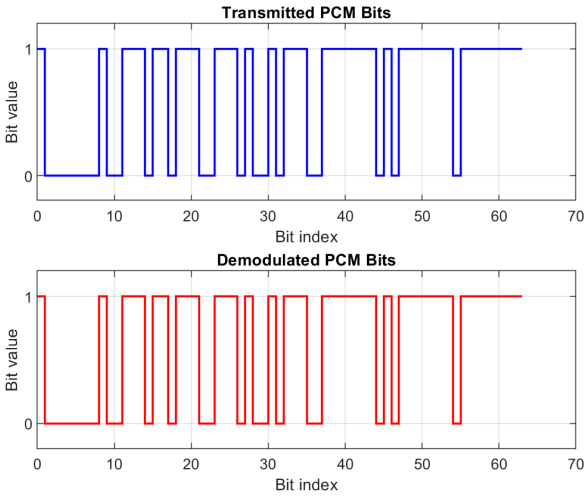


(a) Received waveform through AWGN

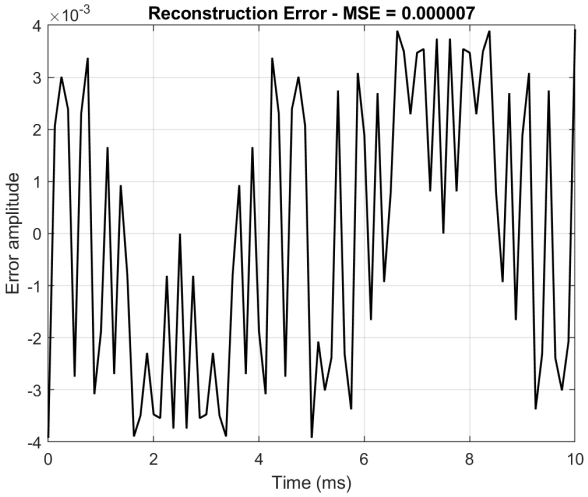


(b) Constellation with noise dispersion

Figure 7: Channel transmission and reception metrics.



(a) Transmitted vs. Demodulated bits



(b) Reconstruction error distribution

Figure 8: Bit recovery and reconstruction error analysis.

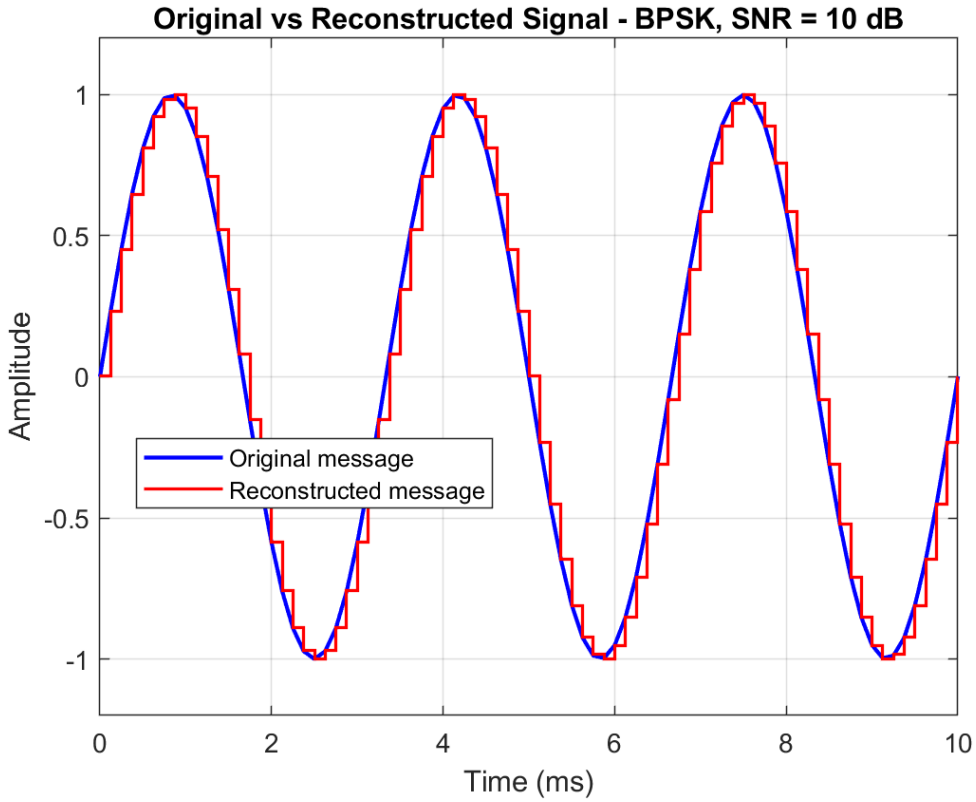


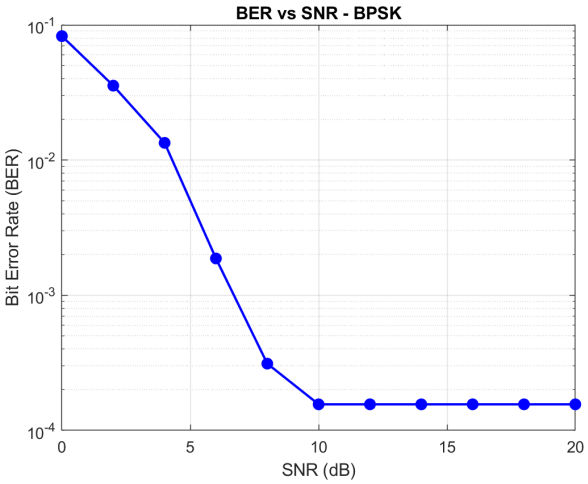
Figure 9: Final comparison between original and reconstructed message signals.

3 Performance Analysis: Channel SNR vs. Reconstruction MSE

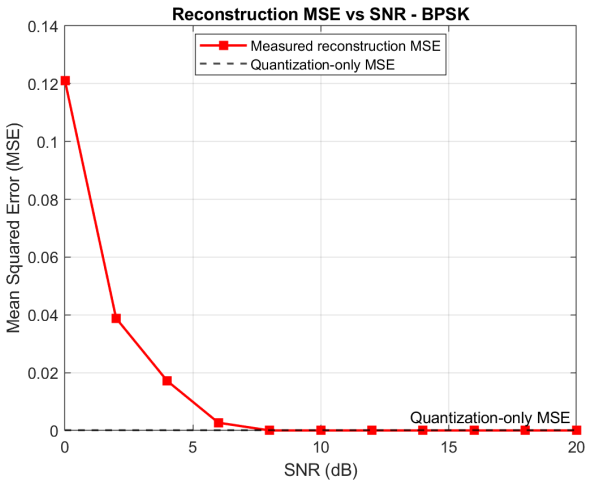
An automated SNR sweep was conducted from 0 dB to 20 dB. The metrics are summarized below:

SNR (dB)	Bit Error Rate (BER)	Reconstruction MSE	Shannon Capacity (kbps)
0.0	0.154925	0.236262	321.78
4.0	0.059227	0.093369	321.78
8.0	0.005923	0.012370	321.78
12.0	0.000000	0.000007	321.78
16.0	0.000000	0.000007	321.78
20.0	0.000000	0.000007	321.78

Table 1: Performance metrics across the evaluated SNR sweep.



(a) Bit Error Rate (BER) vs. SNR



(b) Reconstruction MSE vs. SNR

Figure 10: Bit Error Rate and Reconstruction MSE performance sweeps.

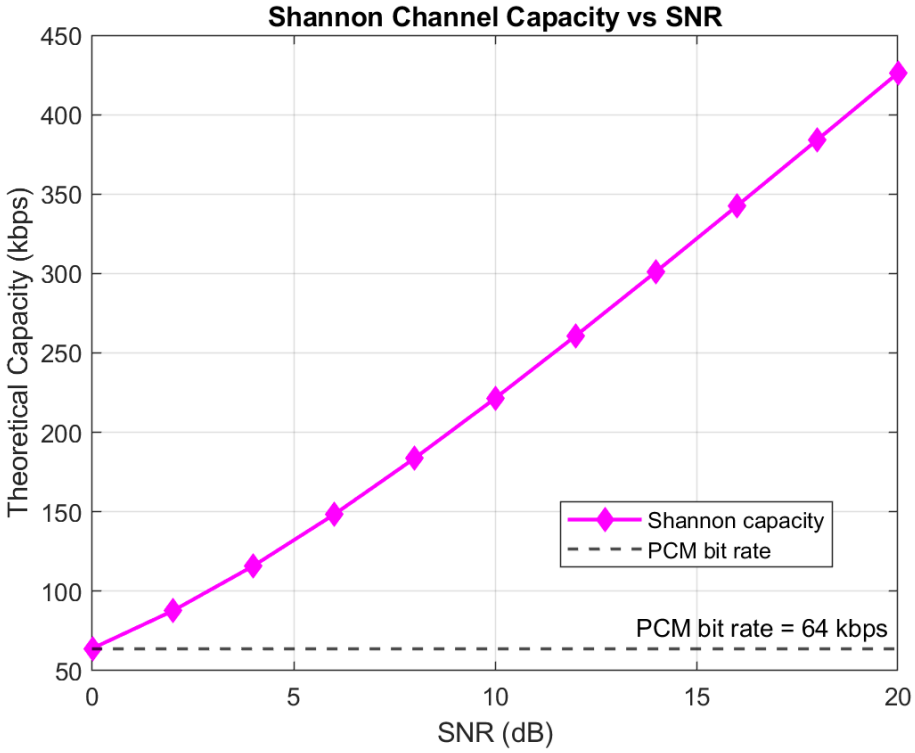


Figure 11: Shannon channel capacity versus SNR compared to the PCM source bit rate.

3.1 Analysis Findings

- **Low SNR Regime (0–6 dB):** High noise variance introduces symbol dispersion, leading to elevated bit error rates and waveform distortion.
- **High SNR Regime ($\geq$ 10 dB):** Constellation points cluster tightly around reference states, driving BER to zero. Reconstruction MSE converges asymptotically to the uniform quantization noise floor ($\Delta^2/12$).

4 Discussion: Shannon's Channel Capacity Theorem

Claude Shannon's Hartley-Shannon theorem defines the theoretical maximum channel capacity C:

$$C = B \log_2 \left(1 + \frac{S}{N}\right) \tag{1}$$

4.1 Bottleneck Evaluation

The PCM source bit rate is $R_b = F_s \times n_{\text{bits}} = 64\,\text{kbps}$. Across the tested SNR range (0–20 dB), channel capacity comfortably exceeds 64 kbps (as seen in Figure 11), confirming sufficient bandwidth headroom. The performance bottleneck shifts from channel capacity limitations to internal quantizer resolution (Δ) at high SNR.

5 Conclusion

The simulator successfully executes all six stages of an end-to-end communication system, aligning with theoretical expectations and quantization noise limits.